

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2019

Mathematics

Paper 2

Higher Level

Time: 2 hours, 30 minutes

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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(Suggested maximum time: 15 minutes)

Round 1														Final Round												
6 5 7																										
8	8	8	7	7	7	6	6	5	5	5	3	0	8	1	2	4	5	5	6	6	7	8	9	9	9	9
				9	9	7	4	4	3	3	2	0	9	0	1	1	2	2	3	3	4	6	9	9		
							6	4	3	2	1	10	0	2	2	4	4	6								

Key: | 9 | 4 = 94 metres

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

(Suggested maximum time: 10 minutes)

(a) Complete the **sample space** below showing all of the possible outcomes. Two have already been filled in. **E E O** means that Nicola recorded two even numbers followed by an odd number when rolling the die.

E E E			

- [illegible]

- Answer:**

Reason:

- [illegible]

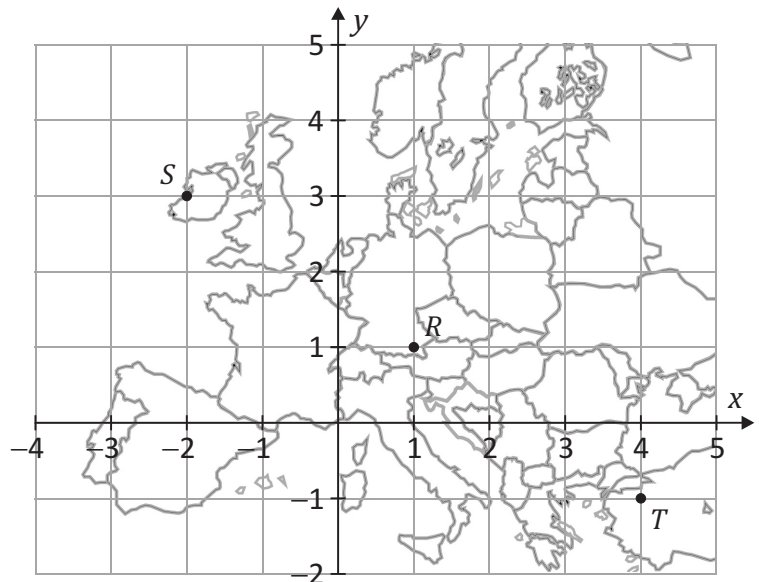
- [illegible]

Question 3

(Suggested maximum time: 10 minutes)

The co-ordinate diagram below shows a map of Europe with airports indicated using the points R , S and T .

- (a)** Write down the co-ordinates of the points which represent the airports R , S and T .

$$R\left(\begin{array}{c} \\ \\ \end{array} , \begin{array}{c} \\ \\ \end{array} \right)$$
$$S(\quad, \quad)$$
$$T(\quad, \quad)$$


- (b)** Use a co-ordinate geometry method to show that airport R is exactly halfway between airports S and T .

[illegible]

- (c) Find $|ST|$, the distance between airports S and T .

[illegible]

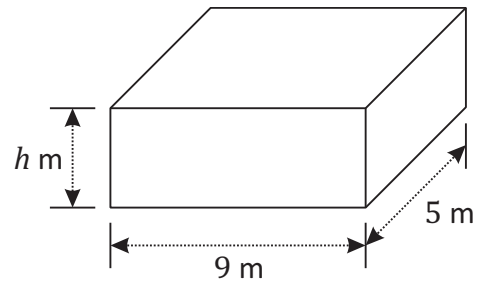
- (d)** The scale on the map is 1 grid unit length = 450 km.
If a plane flies at an average speed of 600 km/h and takes the most direct route, how long would the flight from airport *S* to airport *T* take?
Give your answer correct to the nearest minute.

[illegible]

Question 4

(Suggested maximum time: 15 minutes)

A rectangular water storage tank, open at the top, has internal dimensions 9 m long, 5 m wide and h m high, as shown.



- (a) Show that the **total internal surface area** of the tank is $(28h + 45) \text{ m}^2$.

A grid of 100 squares (10x10) is shown. A scale bar at the top right indicates a length of 9 m, with arrows pointing to the left and right ends of the bar.

- (b)** Given that the total internal surface area of the tank is 143 m^2 , find the value of h .

[illegible]

- (c) Work out the **volume** of the tank in m^3 .

[illegible]

- (d) Water flows into the tank through a cylindrical pipe with an internal radius of 10 cm at a rate of 30 cm/sec. How long will it take to fill the water tank? Give your answer in hours and minutes, correct to the nearest minute.

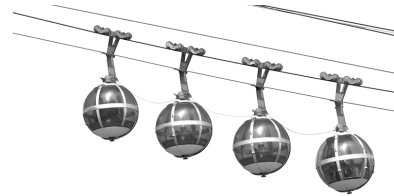
A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 1 cm by 1 cm each. There are 20 columns and 20 rows of squares, creating a total area of 400 small squares. The grid lines are thin and evenly spaced.

Question 5

(Suggested maximum time: 10 minutes)

The Bubbles of Grenoble is a cable car system built in the French city of Grenoble to connect the city centre to the Bastille, a former fortress overlooking the city.

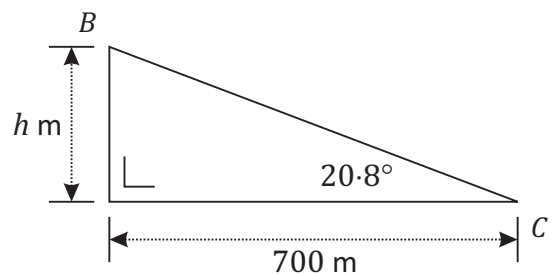
Each cable car is in the shape of a sphere with a radius of 2 metres.



- (a)** Find the **volume** of one of the cable cars, correct to the nearest m^3 .

[illegible]

The diagram shows the journey that the cable cars take from the city centre (C) to the Bastille (B). They travel a **horizontal distance** of 700 metres and at an angle of elevation of 20.8° .



- (b)** Use trigonometry to find h , the **vertical height** that the cable cars climb travelling from C to B . Give your answer correct to the nearest metre.

[illegible]

- (c) Find the **actual distance** that the cable cars travel from C to B .
Give your answer in metres, correct to one decimal place.

[illegible]

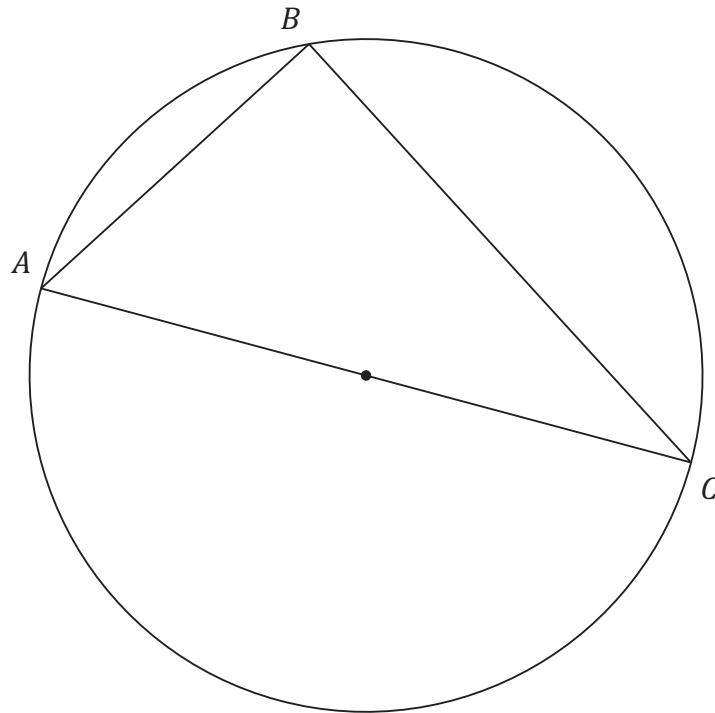
- (d) The cable cars take 3.5 minutes to travel from C to B .
Find the **average speed** of the cable cars in metres per second, correct to one decimal place.

[illegible]

Question 6 (Suggested maximum time: 10 minutes)

Question 6 (Suggested maximum time: 10 minutes)

The diagram shows the triangle ABC inscribed in a circle, where $|AB| = 48$ mm and $|BC| = 75$ mm.



- (a)** AC is a diameter of the circle and therefore $\angle ABC$ is a right angle.
State a corollary or theorem on your course that confirms the above statement.

[illegible]

- (b)** Find the size of the **smallest angle** in the triangle, correct to the nearest degree.

[illegible]

- (c)** In the diagram above, construct a line perpendicular to $|AC|$ that passes through point B . Show all your construction lines clearly.

(Suggested maximum time: 15 minutes)

The equations of each of the lines l , m , n , o , p and q are given in the tables below.

Line	Equation
p	$y = 2x$
q	$x + 3y - 1 = 0$

- (i) Line has the greatest slope.
- (ii) Line has a negative slope.
- (iii) Lines and are perpendicular to each other.
- (iv) Lines and cross the y -axis at the same point.
- (v) A translation can be used to map line onto line .
- (vi) Line passes through the origin.

- [illegible]

- [illegible]

Question 8 (Suggested maximum time: 10 minutes)

Question 8 (Suggested maximum time: 10 minutes)

A local gym would like to run 5 different exercise classes at 3 different levels, twice a day as shown in the table below.

Exercise Classes		
Type	Level	Time
Aerobics	Beginners	Morning
Pilates	Intermediate	Evening
Yoga	Advanced	
Zumba		
Spin		

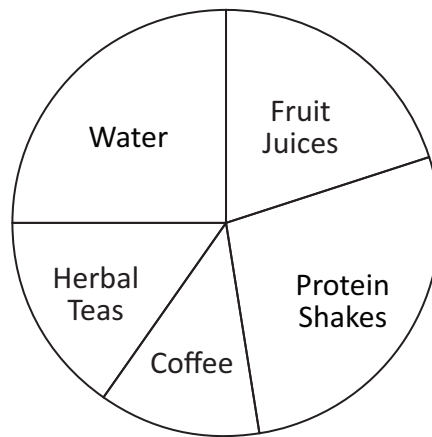
- (a) If every level and time is to be accommodated for each class type, find the total number of classes the gym will need to run.

[illegible]

- (b)** The gym has 2 rooms available for classes.
It can run classes from 7:00 a.m. until 12:00 noon and from 5:00 p.m. until 9:00 p.m.
Each class is 1 hour long.
What percentage of classes can be accommodated?

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

The pie chart below shows the sales of drinks on one particular day.



- (c) Using a protractor, measure the size of each sector in the pie chart and complete the following table, showing the size of each sector and the number of units of each drink sold.

Drink type	Size of angle (degrees)	Number of units sold
Water	90°	40
Fruit Juices		
Protein Shakes		
Coffee		
Herbal Teas		

[illegible]

Question 9

(Suggested maximum time: 20 minutes)

A group of Transition Year students were asked to record the time they spent online between 09:00 on Saturday and 09:00 on Monday.

The table shows the total number of hours that each student in the group spent online over this period.

6.5	9.2	7.5	5	0	7.4	3.4
8.4	8.5	4	6.8	7.8	5.4	4.2
9	3	9.9	2.8	8	6.9	2.9
6.8	4.9	6	5.1	3.6	5.8	8.1

- (a)** Use this data to complete the following frequency table.

Hours online	0–2	2–4	4–6	6–8	8–10
Tally					
Frequency					

[Note: 2–4 means 2 hours or more but less than 4 hours, etc.]

- (b)** Paula says, “There is an outlier in the data.” What value do you think she is referring to? Is she correct? Give a reason for your answer.

[illegible]

- (c) Use **mid-interval values** to estimate the **mean** number of hours spent online by the group. Give your answer correct to two decimal places.

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

- (d)** Draw a **histogram** to represent the data from your frequency table in part **(a)**. Label each axis clearly.

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

- (e)** Do you think these results can be used as a representative sample of the time spent online by students in the entire school? Give a reason for your answer.

[illegible]

- (f) Dylan says, “It is obvious from the histogram that the **modal** time interval was **6–8**.” Do you agree with Dylan? Give a reason for your answer.

Answer:	
Reason:	

Question 10

(Suggested maximum time: 5 minutes)

Prove that each exterior angle of a triangle is equal to the sum of the interior opposite angles.

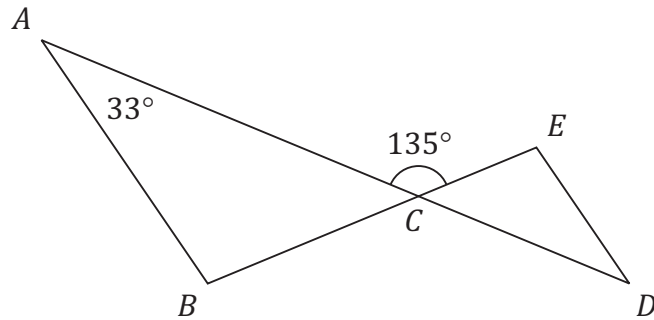
Diagram:

<i>Given:</i>	
<i>To Prove:</i>	
<i>Construction:</i>	
<i>Proof:</i>	

Question 11

(Suggested maximum time: 15 minutes)

In the diagram, $|AB|$ is parallel to $|DE|$.

$$|\angle ACE| = 135^\circ \text{ and } |\angle DAB| = 33^\circ.$$


- (a) Find $|\angle DEB|$.

[illegible]

- (b)** Show that the triangle ABC is similar to the triangle CDE .

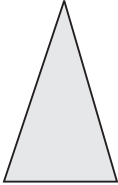
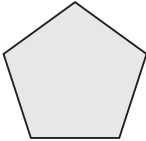
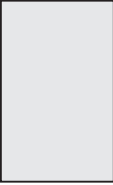
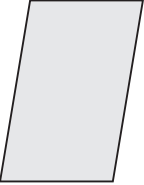
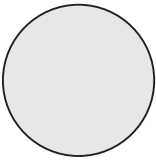
[illegible]

- (c)** $|CE| = 5$ cm, $|AC| = 16$ cm and $|BC| = |CD|$.
Find $|BC|$, giving your answer correct to one decimal place.

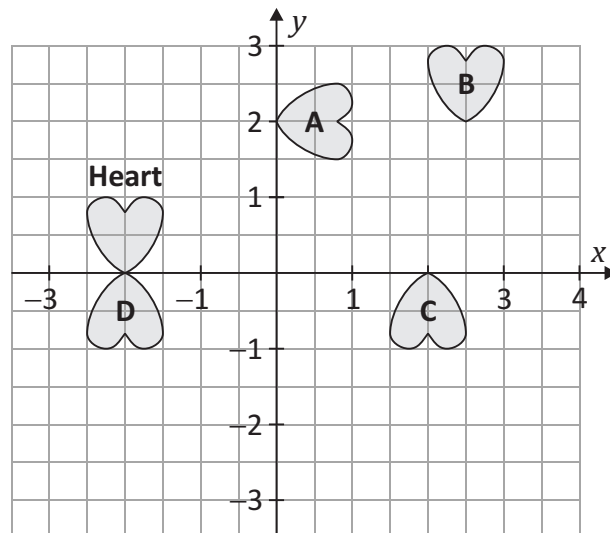
Question 12

(Suggested maximum time: 5 minutes)

- (a) The images and names of shapes are given in the table below.
Write down the number of axes of symmetry of each shape, below the name of the shape, in the box provided.

Image					
Name	Isosceles Triangle	Pentagon	Rectangle	Parallelogram	Circle
Number of axes of symmetry					

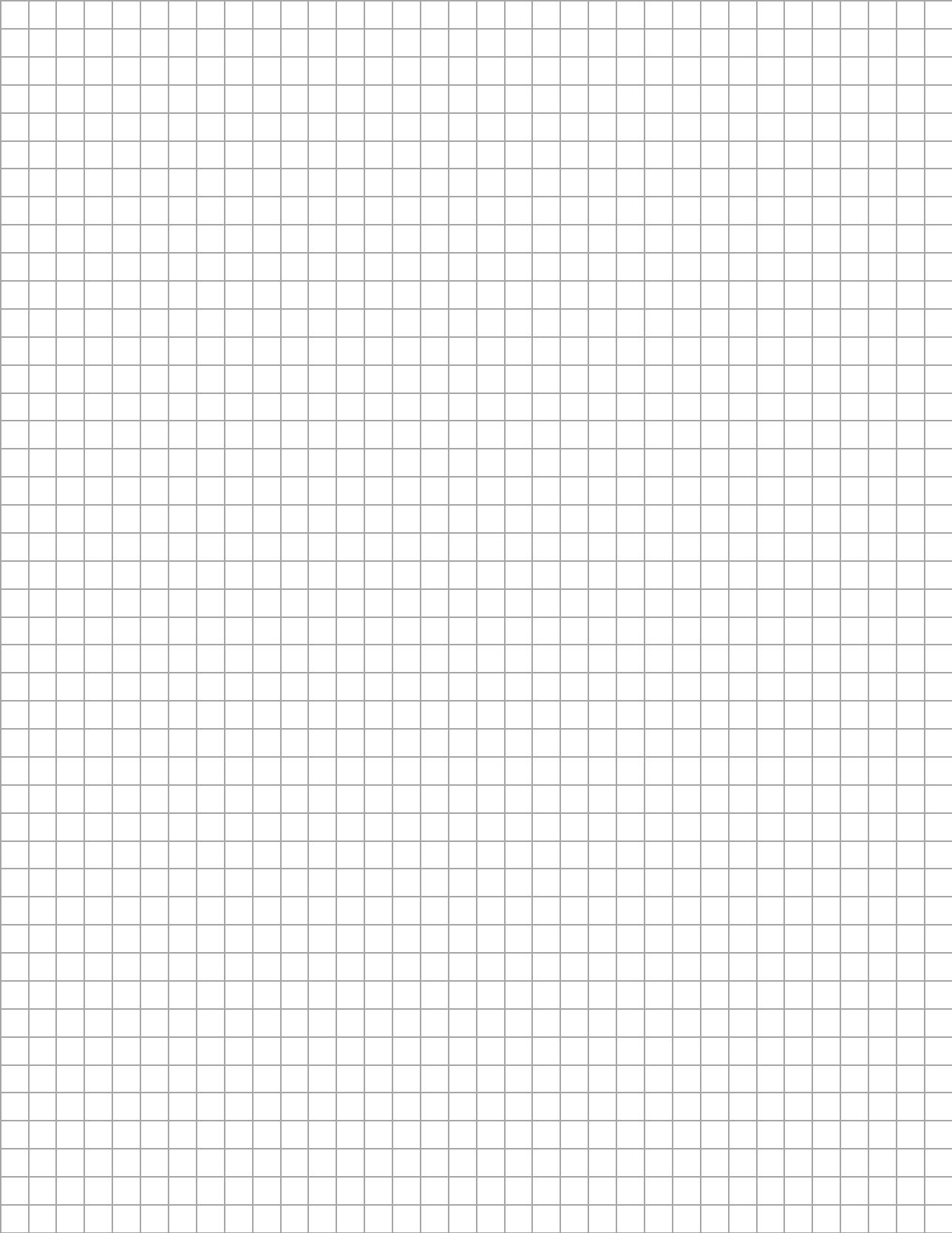
- (b) The co-ordinate grid below shows a heart shape and four images of the shape, **A**, **B**, **C** and **D**, under four different transformations.



Complete the table by writing **A**, **B**, **C** or **D** in the correct place to show which image comes from each transformation. You may use each letter only **once**.

Transformation	Image (A, B, C or D)
Translation	
Axial Symmetry	
Central Symmetry in (0, 0)	
Rotation about (0, 0)	

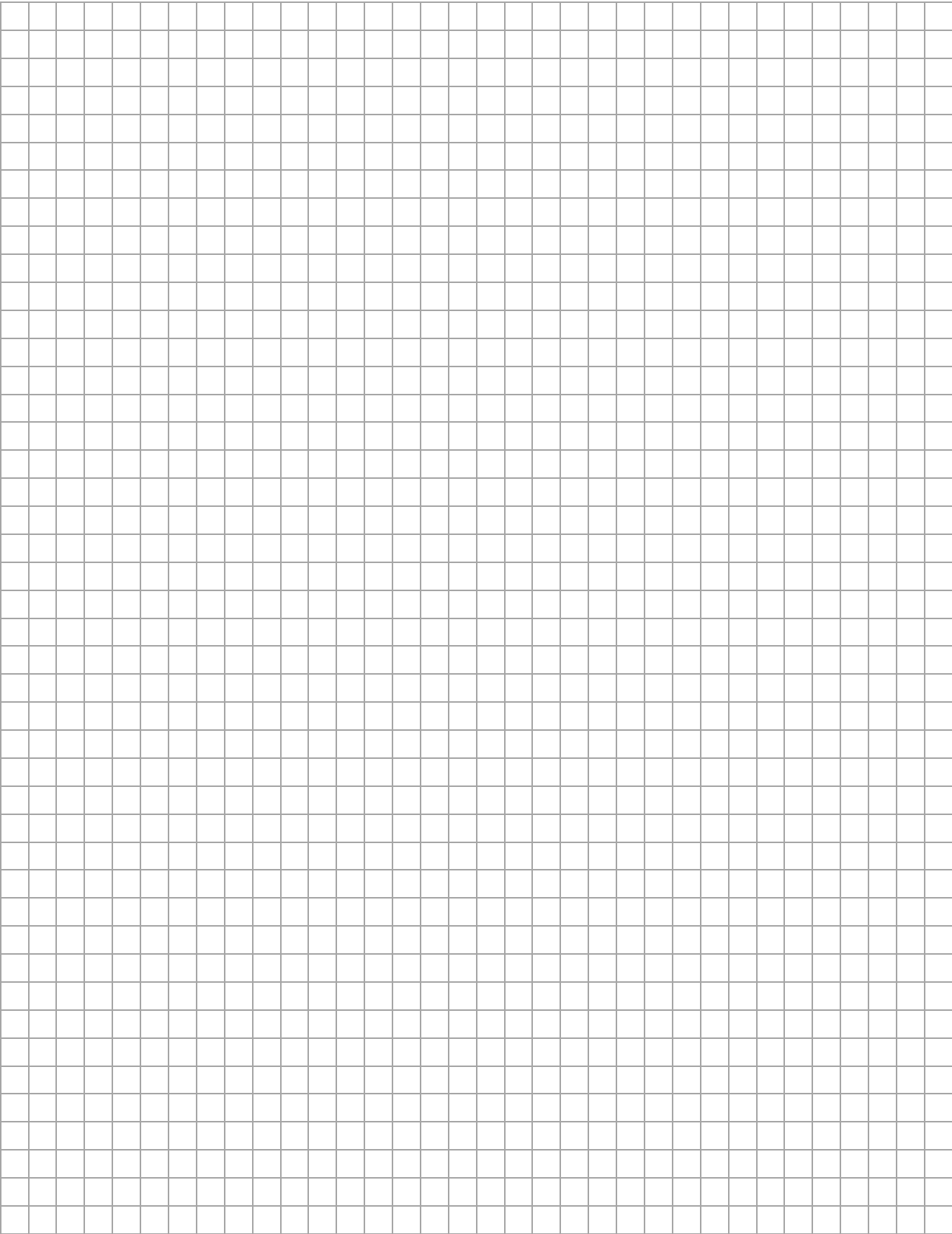
Page for extra work.
Label any extra work clearly with the question number and part.



Label any extra work clearly with the question number and part.

This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

Page for extra work.
Label any extra work clearly with the question number and part.



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This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

Pre-Junior Certificate Examination, 2019

Mathematics – Paper 2

Higher Level

Time: 2 hours, 30 minutes

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